

LAB DAY 4

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COURSE CODE AND NAME: CSA0617 DESIGN AND ANALYSIS OF ALGORITHM

DICE THROW PROBLEM

Q1. 3 DICE, 6 FACES, TARGET=8:

```
def dice_ways(n, m, target):
```

```
    dp = [[0] * (target + 1) for _ in range(n + 1)]
```

```
    dp[0][0] = 1
```

```
    for dice in range(1, n + 1):
```

```
        for total in range(1, target + 1):
```

```
            for face in range(1, m + 1):
```

```
                if total >= face:
```

```
                    dp[dice][total] += dp[dice - 1][total - face]
```

```
    return dp[n][target]
```

```
n = 3
```

```
m = 6
```

```
target = 8
```

```
print("Number of ways =", dice_ways(n, m, target))
```

EXECUTION & OUTPUT:

Q3. 4 DICE, 4 FACES, TAGRET=10:

```

def dice_ways(n, m, target):

    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):        for
j in range(1, target + 1):
    for face in range(1, m + 1):
    if j >= face:

        dp[i][j] += dp[i - 1][j - face]

    return dp[n][target]

print("Number of ways =", dice_ways(4, 4, 10))

```

EXECUTION & OUTPUT:

The screenshot shows a Python IDE with two panes. The left pane contains the code for the `dice_ways` function and its execution. The right pane shows the output of the program, which is "Number of ways = 44".

Q4. 3 DICE, 6 FACES, TARGET=12:

```

def dice_ways(n, m, target):

    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):        for
j in range(1, target + 1):

```

```
for face in range(1, m + 1):
```

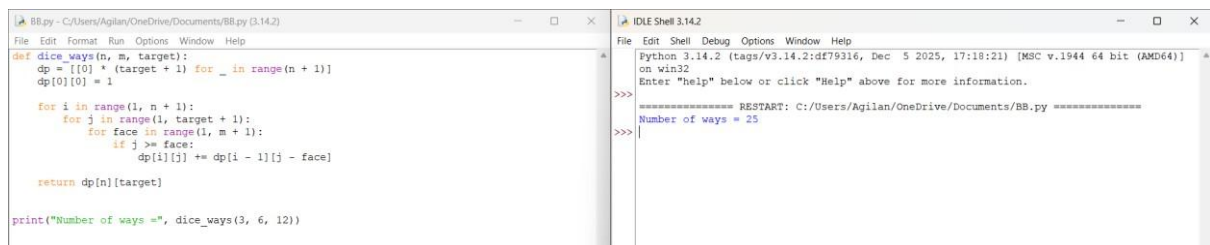
```
if j >= face:
```

```
    dp[i][j] += dp[i - 1][j - face]
```

```
return dp[n][target]
```

```
print("Number of ways =", dice_ways(3, 6, 12))
```

EXECUTION & OUTPUT:



```
BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)
File Edit Format Run Options Window Help
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for j in range(1, target + 1):
            for face in range(1, m + 1):
                if j >= face:
                    dp[i][j] += dp[i - 1][j - face]

    return dp[n][target]

print("Number of ways =", dice_ways(3, 6, 12))

IDLE Shell 3.14.2
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
>>>
Number of ways = 25
>>>
```

Q5. 5 DICE, 6 FACES, TARGET=15:

```
def dice_ways(n, m, target):
```

```
    dp = [[0] * (target + 1) for _ in range(n + 1)]
```

```
    dp[0][0] = 1
```

```
    for i in range(1, n + 1):
```

```
        for j in range(1, target + 1):
```

```
            for face in range(1, m + 1):
```

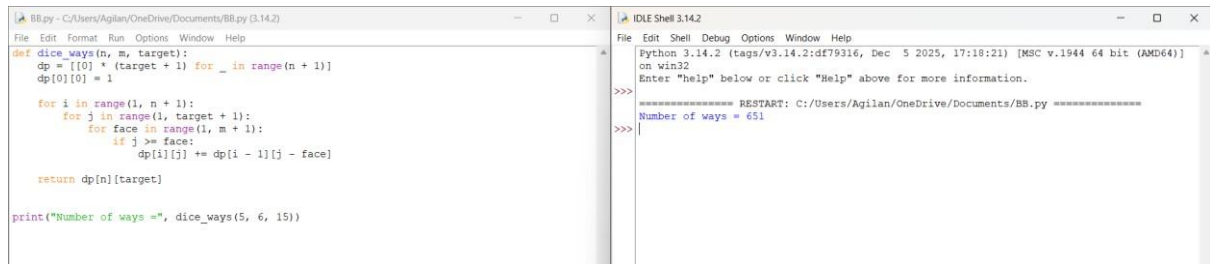
```
                if j >= face:
```

```
                    dp[i][j] += dp[i - 1][j - face]
```

```
return dp[n][target]
```

```
print("Number of ways =", dice_ways(5, 6, 15))
```

EXECUTION & OUTPUT:



The screenshot shows two windows. The left window, titled 'BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)', contains the following Python code:

```
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for j in range(1, target + 1):
            for face in range(1, m + 1):
                if j >= face:
                    dp[i][j] += dp[i - 1][j - face]

    return dp[n][target]

print("Number of ways =", dice_ways(5, 6, 15))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
Number of ways = 651
>>>
```

Q6. 3 DICE, 5 FACES, TARGET=9:

```
def dice_ways(n, m, target):
```

```
    dp = [[0] * (target + 1) for _ in range(n + 1)]
```

```
    dp[0][0] = 1
```

```
        for i in range(1, n + 1):
```

```
            for j in range(1, target + 1):
```

```
                for face in range(1, m + 1):
```

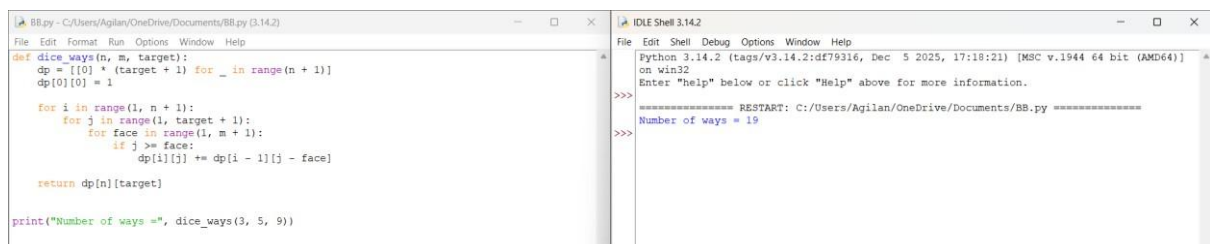
```
                    if j >= face:
```

```
                        dp[i][j] += dp[i - 1][j - face]
```

```
    return dp[n][target]
```

```
print("Number of ways =", dice_ways(3, 5, 9))
```

EXECUTION & OUTPUT:



The screenshot shows two windows. The left window, titled 'BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)', contains the following Python code:

```
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for j in range(1, target + 1):
            for face in range(1, m + 1):
                if j >= face:
                    dp[i][j] += dp[i - 1][j - face]

    return dp[n][target]

print("Number of ways =", dice_ways(3, 5, 9))
```

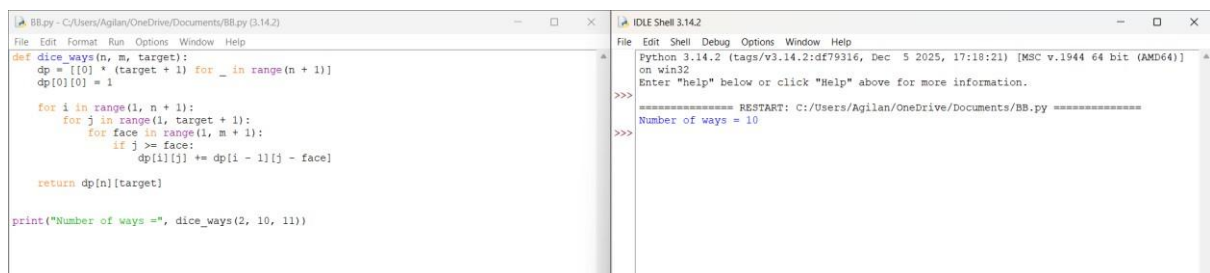
The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
Number of ways = 19
>>>
```

Q7. 2 DICE, 10 FACES, TARGET=11:

```
def dice_ways(n, m, target):  
    dp = [[0] * (target + 1) for _ in range(n + 1)]  
    dp[0][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, target + 1):  
            for face in range(1, m + 1):  
                if j >= face:  
                    dp[i][j] += dp[i - 1][j - face]  
  
    return dp[n][target]  
  
print("Number of ways =", dice_ways(2, 10, 11))
```

EXECUTION & OUTPUT:



```
BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)  
File Edit Format Run Options Window Help  
def dice_ways(n, m, target):  
    dp = [[0] * (target + 1) for _ in range(n + 1)]  
    dp[0][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, target + 1):  
            for face in range(1, m + 1):  
                if j >= face:  
                    dp[i][j] += dp[i - 1][j - face]  
  
    return dp[n][target]  
  
print("Number of ways =", dice_ways(2, 10, 11))  
  
IDLE Shell 3.14.2  
File Edit Shell Debug Options Window Help  
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====  
>>> Number of ways = 10  
>>>
```

Q8. GNEREAL N DICE, M FACES:

```
def dice_ways(n, m, target):  
    dp = [[0] * (target + 1) for _ in range(n + 1)]  
    dp[0][0] = 1
```

```

        for i in range(1, n + 1):            for
total in range(1, target + 1):
for face in range(1, m + 1):
if total >= face:

    dp[i][total] += dp[i - 1][total - face]

return dp[n][target]

```

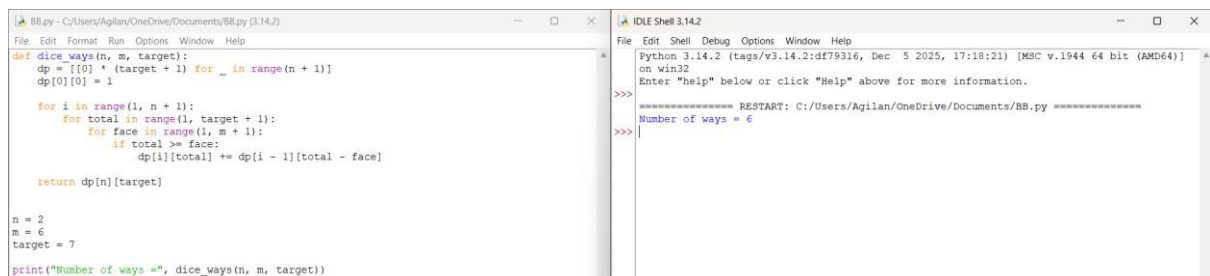
n = 2

m = 6

target = 7

```
print("Number of ways =", dice_ways(n, m, target))
```

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window displays the code for the dice_ways function and its execution with n=2, m=6, and target=7. The right window shows the output of the program, which is 'Number of ways = 6'.

```

BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)
File Edit Format Run Options Window Help
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for total in range(1, target + 1):
            for face in range(1, m + 1):
                if total >= face:
                    dp[i][total] += dp[i - 1][total - face]

    return dp[n][target]

n = 2
m = 6
target = 7
print("Number of ways =", dice_ways(n, m, target))

IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
>>>
Number of ways = 6
>>>

```

Q9. 3 DICE, 6 FACES, TARGET=8:

```
def dice_ways(n, m, target):
```

```
    dp = [[0] * (target + 1) for _ in range(n + 1)]
```

```
    dp[0][0] = 1
```

```

        for i in range(1, n + 1):            for
total in range(1, target + 1):

```

```

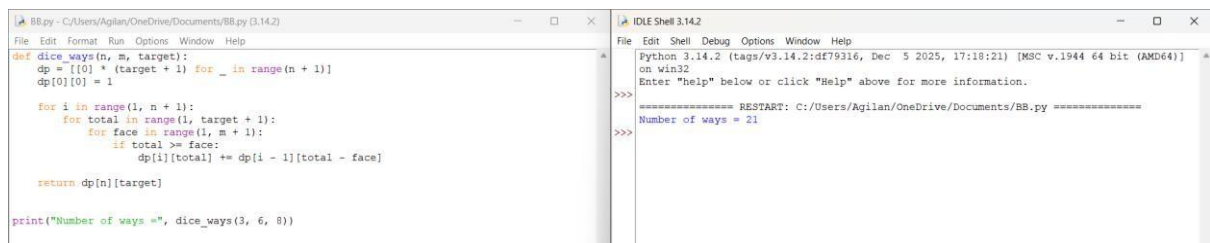
for face in range(1, m + 1):
    if total >= face:
        dp[i][total] += dp[i - 1][total - face]

return dp[n][target]

print("Number of ways =", dice_ways(3, 6, 8))

```

EXECUTION & OUTPUT:



```

BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)
File Edit Format Run Options Window Help
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for total in range(1, target + 1):
            for face in range(1, m + 1):
                if total >= face:
                    dp[i][total] += dp[i - 1][total - face]

    return dp[n][target]

print("Number of ways =", dice_ways(3, 6, 8))

IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
>>>
Number of ways = 21
>>>

```

Q10. 4 DICE, 6 FACES, TARGET=10:

```

def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

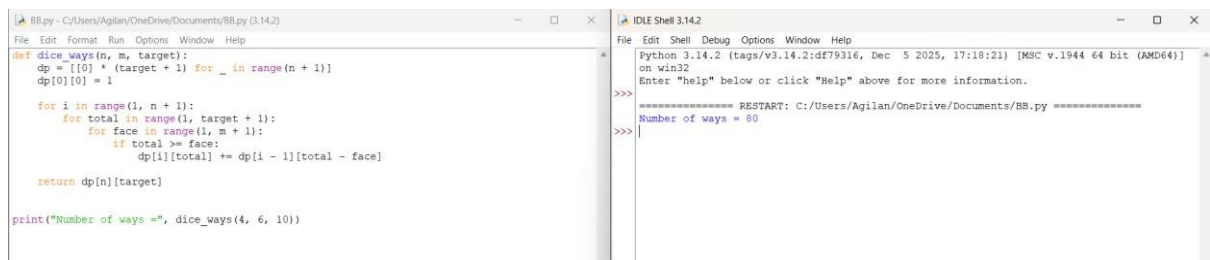
    for i in range(1, n + 1):
        for total in range(1, target + 1):
            for face in range(1, m + 1):
                if total >= face:
                    dp[i][total] += dp[i - 1][total - face]

    return dp[n][target]

print("Number of ways =", dice_ways(4, 6, 10))

```

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)', contains the following code:

```
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for total in range(1, target + 1):
            for face in range(1, m + 1):
                if total >= face:
                    dp[i][total] += dp[i - 1][total - face]

    return dp[n][target]

print("Number of ways =", dice_ways(4, 6, 10))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
Number of ways = 80
>>>
```

Q11. 2 DICE, 4 FACES, TARGET=5:

def dice_ways(n, m, target):

 dp = [[0] * (target + 1) for _ in range(n + 1)]

 dp[0][0] = 1

 for i in range(1, n + 1):

 for total in range(1, target + 1):

 for face in range(1, m + 1):

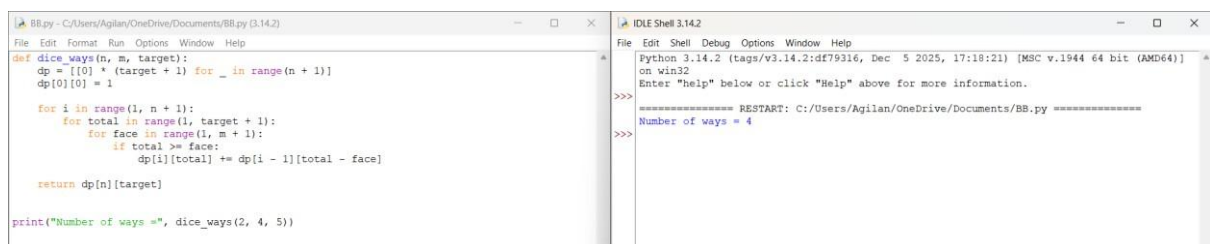
 if total >= face:

 dp[i][total] += dp[i - 1][total - face]

 return dp[n][target]

print("Number of ways =", dice_ways(2, 4, 5))

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)', contains the following code:

```
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for total in range(1, target + 1):
            for face in range(1, m + 1):
                if total >= face:
                    dp[i][total] += dp[i - 1][total - face]

    return dp[n][target]

print("Number of ways =", dice_ways(2, 4, 5))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
Number of ways = 4
>>>
```

BINOMIAL COEFFICIENT

DICE THROWS-3 DICE, 4 FACES, TARGET=7:

```
def dice_ways(n, m, target):
```

```
    dp = [[0] * (target + 1) for _ in range(n + 1)]
```

```
    dp[0][0] = 1
```

```
    for i in range(1, n + 1):        for
```

```
        j in range(1, target + 1):
```

```
            for face in range(1, m + 1):
```

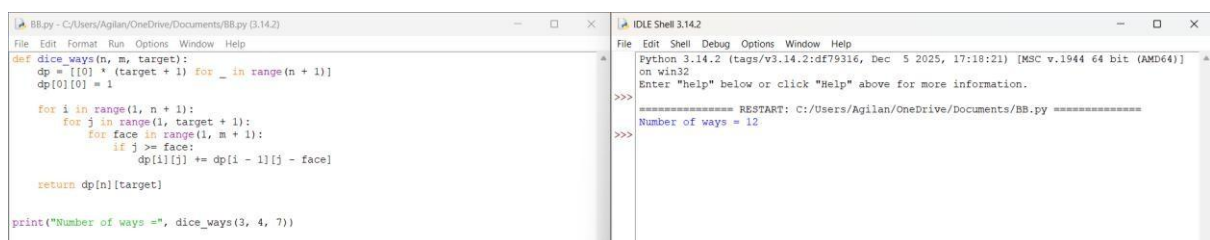
```
                if j >= face:
```

```
                    dp[i][j] += dp[i - 1][j - face]
```

```
    return dp[n][target]
```

```
print("Number of ways =", dice_ways(3, 4, 7))
```

EXECUTION & OUTPUT:



```
BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)
File Edit Format Run Options Window Help
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for j in range(1, target + 1):
            for face in range(1, m + 1):
                if j >= face:
                    dp[i][j] += dp[i - 1][j - face]

    return dp[n][target]

print("Number of ways =", dice_ways(3, 4, 7))

IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
>>> Number of ways = 12
>>>
```

Q2. C(5,2):

```
def binomial(n, r):
```

```
    dp = [[0] * (r + 1) for _ in range(n + 1)]
```

```
    for i in range(n + 1):
```

```
        dp[i][0] = 1
```

```

        for i in range(1, n + 1):
            for
j in range(1, min(i, r) + 1):

                dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

return dp[n][r]

```

```
print("C(5,2) =", binomial(5, 2))
```

EXECUTION & OUTPUT:

The screenshot shows a Python IDE with two panes. The left pane displays the code for the binomial function, and the right pane shows the output of the program.

```

def binomial(n, r):
    dp = [[0] * (r + 1) for _ in range(n + 1)]

    for i in range(n + 1):
        dp[i][0] = 1

    for i in range(1, n + 1):
        for j in range(1, min(i, r) + 1):
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

    return dp[n][r]

print("C(5,2) =", binomial(5, 2))

```

The output in the right pane is:

```

>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====
C(5,2) = 10
>>>

```

C(8,3):

```

def binomial(n, r):

    dp = [[0] * (r + 1) for _ in range(n + 1)]

    for i in range(n + 1):

        dp[i][0] = 1

    for i in range(1, n + 1):
        for
j in range(1, min(i, r) + 1):

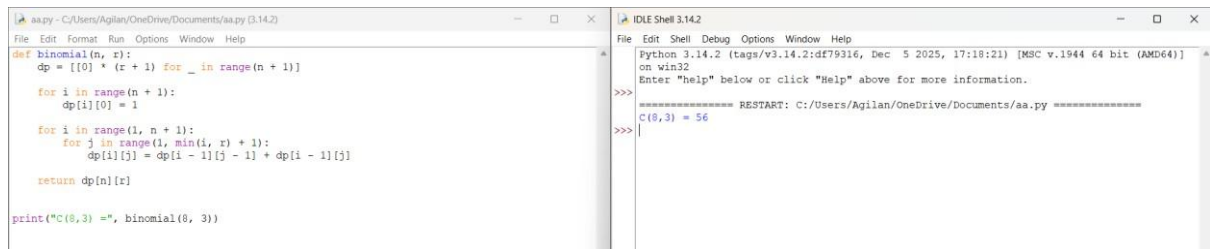
                dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

return dp[n][r]

```

```
print("C(8,3) =", binomial(8, 3))
```

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'aa.py - C:/Users/Agilan/OneDrive/Documents/aa.py (3.14.2)', contains the following code:

```
def binomial(n, r):  
    dp = [[0] * (r + 1) for _ in range(n + 1)]  
  
    for i in range(n + 1):  
        dp[i][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, min(i, r) + 1):  
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]  
  
    return dp[n][r]  
  
print("C(8,3) =", binomial(8, 3))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
>>> Enter "help" below or click "Help" above for more information.  
>>> ===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====  
>>> C(8,3) = 56  
>>>
```

C(10,4):

def binomial(n, r):

dp = [[0] * (r + 1) for _ in range(n + 1)]

for i in range(n + 1):

dp[i][0] = 1

for i in range(1, n + 1): for

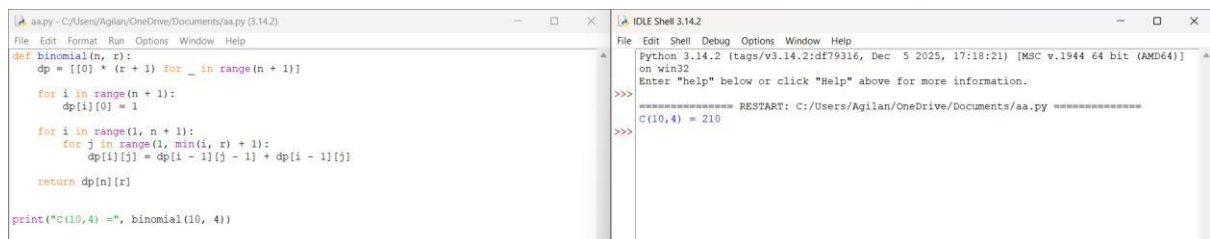
j in range(1, min(i, r) + 1):

dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

return dp[n][r]

print("C(10,4) =", binomial(10, 4))

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'aa.py - C:/Users/Agilan/OneDrive/Documents/aa.py (3.14.2)', contains the following code:

```
def binomial(n, r):  
    dp = [[0] * (r + 1) for _ in range(n + 1)]  
  
    for i in range(n + 1):  
        dp[i][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, min(i, r) + 1):  
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]  
  
    return dp[n][r]  
  
print("C(10,4) =", binomial(10, 4))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
>>> Enter "help" below or click "Help" above for more information.  
>>> ===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====  
>>> C(10,4) = 210  
>>>
```

C(12,5):

def binomial(n, r):

```
dp = [[0] * (r + 1) for _ in range(n + 1)]
```

```
for i in range(n + 1):
```

```
    dp[i][0] = 1
```

```
    for i in range(1, n + 1):
```

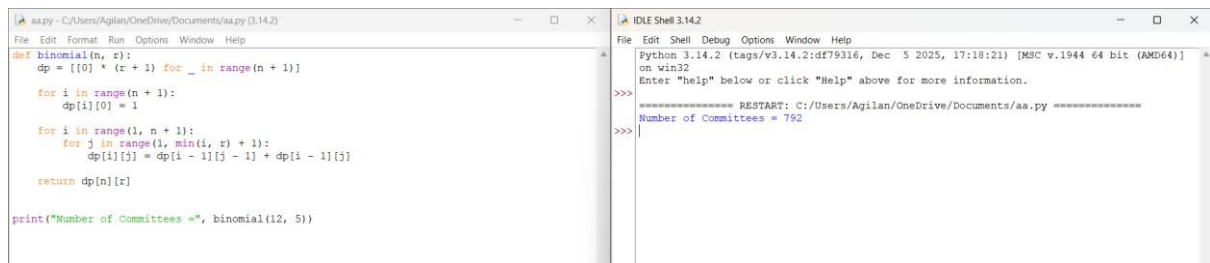
```
        for j in range(1, min(i, r) + 1):
```

```
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
```

```
    return dp[n][r]
```

```
print("Number of Committees =", binomial(12, 5))
```

EXECUTION & OUTPUT:



```
def binomial(n, r):
    dp = [[0] * (r + 1) for _ in range(n + 1)]

    for i in range(n + 1):
        dp[i][0] = 1

    for i in range(1, n + 1):
        for j in range(1, min(i, r) + 1):
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

    return dp[n][r]

print("Number of Committees =", binomial(12, 5))
```

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====
Number of Committees = 792
>>>
```

C(15,6):

```
def binomial(n, r):
```

```
    dp = [[0] * (r + 1) for _ in range(n + 1)]
```

```
    for i in range(n + 1):
```

```
        dp[i][0] = 1
```

```
    for i in range(1, n + 1):
```

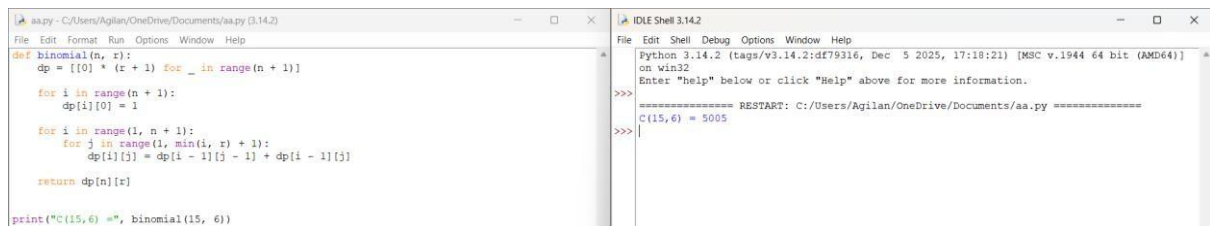
```
        for j in range(1, min(i, r) + 1):
```

```
dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
```

```
return dp[n][r]
```

```
print("C(15,6) =", binomial(15, 6))
```

EXECUTION & OUTPUT:

A screenshot of a Python IDE window titled 'aa.py - C:/Users/Agilan/OneDrive/Documents/aa.py (3.14.2)'. The code defines a binomial function using dynamic programming. The output window on the right shows the execution of the function with arguments (15, 6), resulting in the value 5005.

```
def binomial(n, r):  
    dp = [[0] * (r + 1) for _ in range(n + 1)]  
  
    for i in range(n + 1):  
        dp[i][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, min(i, r) + 1):  
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]  
  
    return dp[n][r]  
  
print("C(15,6) =", binomial(15, 6))
```

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>> ===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====  
>>> C(15,6) = 5005  
>>> |
```

C(10,3):

```
def binomial(n, r):
```

```
    dp = [[0] * (r + 1) for _ in range(n + 1)]
```

```
    for i in range(n + 1):
```

```
        dp[i][0] = 1
```

```
        for i in range(1, n + 1):
```

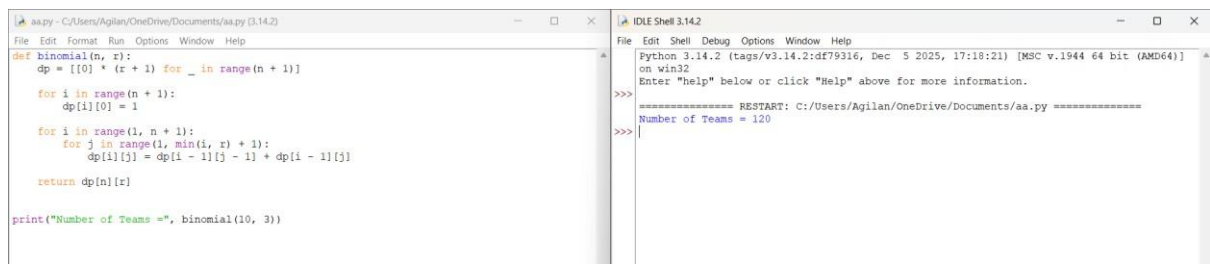
```
            for j in range(1, min(i, r) + 1):
```

```
                dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
```

```
    return dp[n][r]
```

```
print("Number of Teams =", binomial(10, 3))
```

EXECUTION & OUTPUT:



```
def binomial(n, r):
    dp = [[0] * (r + 1) for _ in range(n + 1)]

    for i in range(n + 1):
        dp[i][0] = 1

    for i in range(1, n + 1):
        for j in range(1, min(i, r) + 1):
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

    return dp[n][r]

print("Number of Teams =", binomial(10, 3))
```

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====
Number of Teams = 120
>>>
```

$C(12,4)$:

def binomial(n, r):

dp = [[0] * (r + 1) for _ in range(n + 1)]

for i in range(n + 1):

dp[i][0] = 1

for i in range(1, n + 1): for

j in range(1, min(i, r) + 1):

dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

return dp[n][r]

print("Number of Committees =", binomial(12, 4))

EXECUTION & OUTPUT:



```
def binomial(n, r):
    dp = [[0] * (r + 1) for _ in range(n + 1)]

    for i in range(n + 1):
        dp[i][0] = 1

    for i in range(1, n + 1):
        for j in range(1, min(i, r) + 1):
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

    return dp[n][r]

print("Number of Committees =", binomial(12, 4))
```

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====
Number of Committees = 495
>>>
```

$C(15,5)$:

```
def binomial(n, r):
```

```
    dp = [[0] * (r + 1) for _ in range(n + 1)]
```

```
    for i in range(n + 1):
```

```
        dp[i][0] = 1
```

```
        for i in range(1, n + 1):
```

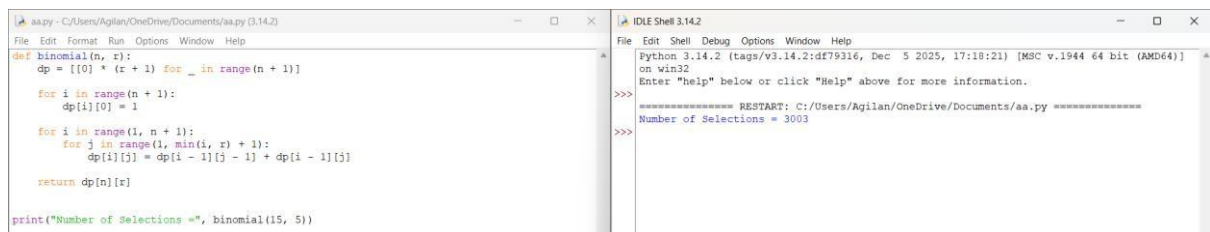
```
            for j in range(1, min(i, r) + 1):
```

```
                dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
```

```
    return dp[n][r]
```

```
print("Number of Selections =", binomial(15, 5))
```

EXECUTION & OUTPUT:

The screenshot shows a Python IDE window with two panes. The left pane displays the code for the binomial coefficient function, including the initialization of the dp array, the base case, the nested loops for calculating values, and the final print statement. The right pane shows the output of the program, which is 'Number of Selections = 3003'. The IDE title bar indicates it is Python 3.14.2.

```
aa.py - C:/Users/Agilan/OneDrive/Documents/aa.py (3.14.2)
File Edit Format Run Options Window Help
def binomial(n, r):
    dp = [[0] * (r + 1) for _ in range(n + 1)]

    for i in range(n + 1):
        dp[i][0] = 1

    for i in range(1, n + 1):
        for j in range(1, min(i, r) + 1):
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

    return dp[n][r]

print("Number of Selections =", binomial(15, 5))

IDLE Shell 3.14.2
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====
>>>
Number of Selections = 3003
```

$C(20,6)$:

```
def binomial(n, r):
```

```
    dp = [[0] * (r + 1) for _ in range(n + 1)]
```

```
    for i in range(n + 1):
```

```
        dp[i][0] = 1
```

```

        for i in range(1, n + 1):
            for
j in range(1, min(i, r) + 1):

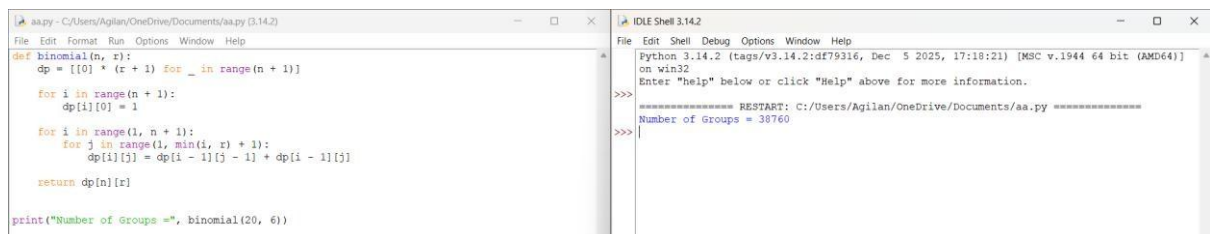
                dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

return dp[n][r]

print("Number of Groups =", binomial(20, 6))

```

EXECUTION & OUTPUT:



$C(8,3)$:

```

def binomial(n, r):

    dp = [[0] * (r + 1) for _ in range(n + 1)]

    for i in range(n + 1):

        dp[i][0] = 1

        for i in range(1, n + 1):
            for
j in range(1, min(i, r) + 1):

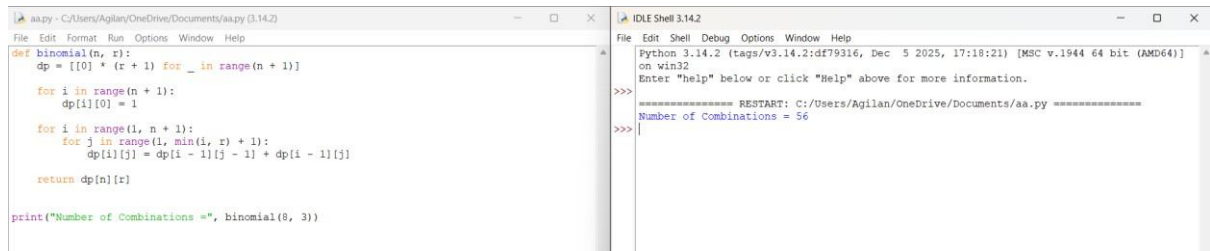
                dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

return dp[n][r]

```

```
print("Number of Combinations =", binomial(8, 3))
```

EXECUTION & OUTPUT:

The screenshot shows a Python IDE with two windows. The left window, titled 'aa.py - C:/Users/Agilan/OneDrive/Documents/aa.py (3.14.2)', contains the following code:

```
def binomial(n, r):  
    dp = [[0] * (r + 1) for _ in range(n + 1)]  
    for i in range(n + 1):  
        dp[i][0] = 1  
    for i in range(1, n + 1):  
        for j in range(1, min(i, r) + 1):  
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]  
    return dp[n][r]  
  
print("Number of Combinations =", binomial(8, 3))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====  
Number of Combinations = 56  
>>>
```

ASSEMBLY LINE SCHEDULING

```
entry = [10, 12] exit
```

```
= [18, 7]
```

```
line1 = [4, 5, 3, 2] line2
```

```
= [2, 10, 1, 4] transfer1
```

```
= [7, 4, 5] transfer2 =
```

```
[9, 2, 8]
```

```
result = assembly_line( entry, exit, line1,
```

```
line2, transfer1, transfer2
```

```
)
```

```
print("Minimum Production Time =", result)
```

EXECUTION & OUTPUT:

```
AA.py - C:\Users\Agilan\OneDrive\Documents\AA.py (3.14.2)
File Edit Format Run Options Window Help
entry = [10, 12]
exit = [18, 7]

line1 = [4, 5, 3, 2]
line2 = [2, 10, 1, 4]

transfer1 = [7, 4, 5]
transfer2 = [9, 2, 8]

result = assembly_line(
    entry, exit, line1, line2, transfer1, transfer2
)

print("Minimum Production Time =", result)
|
```